

How to use this

- Tiers = sequence (start at Tier 0 and climb).
- Modules = group by discipline.
- Core → Depth → References within each module.

Tier 0 — Orientation & Macro Mindset

Core

- Gliner — Global Macro Trading
- Netto — Global macro edge - maximizing return per unit of risk
- Pedersen — Efficiently Inefficient_ How Smart Money Invests and Market Prices Are Determined
- Imanien — Expected Returns
- Imanien — Investing Amid Low Expected Returns
- Harvey, Raltray, Van Hemert — Strategic Risk Management_ Designing Portfolios and Managing Risk
- Macrosynergy — Macro Quantamental Investment Handbook
- Yardeni — Predicting the Markets_ A Professional Autobiography

Depth

- Marks — The Most Important Thing Illuminated_ Uncommon Sense for the Thoughtful Investor
- Marks — Mastering the Market Cycle_ Getting the Odds on Your Side
- Schwager — Market Wizards (series)
- Top Traders Unplugged — Ultimate Guide

References

- Abner — The ETF Handbook
- Dimson, Marsh, Staunton — Triumph of the Optimists_ 101 Years of Global Investment Returns

Tier 1 — Macro Economics & Policy Backbone

Core

- Abel, Bernanke, Croushore — Macroeconomics, Global Edition
- Ljungqvist, Sargent — Recursive Macroeconomic Theory
- Binkler — A Monetary and Fiscal History of the United States, 1961-2021
- Bernanke — 21st Century Monetary Policy_ The Federal Reserve from the Great Inflation to COVID
- Friedman, Schwartz — A Monetary History of the United States, 1867-1960
- Wang — Central Banking 101

Depth

- Leamer — Macroeconomic Patterns and Stories
- Mayes, Siklos, Sturm — The Oxford Handbook of the Economics of Central Banking
- Shiller — Narrative Economics_ How Stories Go Viral and Drive Major Economic Events
- Deacon, Derry, Mirfendereski — Inflation-indexed Securitles
- James, Leister, Rieger — Inflation-Linked Bonds and Derivatives_ Investing, hedging and valuation principles for practitioners

Geo/Structural context

- Dalo — Principles for Dealing with the Changing World Order_ Why Nations Succeed and Fail
- Acemoglu, Robinson — Why Nations Fail_ The Origins of Power, Prosperity and Poverty
- Acemoglu, Johnson — Power and Progress_ Our Thousand-Year Struggle Over Technology and Prosperity
- Piketty — Capital in the Twenty-First Century
- Rajan — Fault Lines_ How Hidden Fractures Still Threaten the World Economy
- Reinhart, Rogoff — This Time Is Different_ Eight Centuries of Financial Folly
- Schularick — Leveraged_ The New Economics of Debt and Financial Fragility
- Sharma — The Rise and Fall of Nations_ Forces of Change in the Post-Crisis World
- Delong — Slouching towards Utopia an economic history of the twentieth century
- Kennedy — The Rise and Fall of the Great Powers_ Economic Change and Military Conflict from 1500-2000
- Frankopan — The New Silk Roads_ The Present and Future of the World

China/Emerging

- The Handbook of China's Financial System — Amstad, Sun, Xiong
- Pettis — The Volatility Machine_ Emerging Economics and the Threat of Financial Collapse
- Pettis — The Great Rebalancing_ Trade, Conflict, and the Perilous Road Ahead for the World Economy
- Klein, Michael Pettis — Trade Wars Are Class Wars_ ... Peace
- Economy — The Third Revolution_ Xi Jinping and the New Chinese State
- Saich — Governance and Politics of China
- Vogel — Deng Xiaoping and the Transformation of China
- Vogel, Jia — The Golden Age of the U.S.-China-Japan Triangle, 1972-1989
- Ezra F. Vogel — Japan as Number One_ Lessons for America-Harvard University Press

Tier 2 — Time Series, Econometrics & Forecasting

Core

- Hamilton — Time Series Analysis
- Tsay — Analysis of Financial Time Series
- Shumway, Stoffer — Time Series Analysis and Its Applications
- Durbin, Koopman — Time Series Analysis by State Space Methods
- Athanasopoulos, Hyndman — Forecasting Principles and Practice
- Wooldridge — Econometric Analysis of Cross Section and Panel Data
- Greene — Econometric Analysis
- Sheppard — financial-econometrics-2020-2021

Depth

- Fuller — Introduction to Statistical Time Series
- Harvey — Time Series Models_ 2nd Edition
- Poon — A Practical Guide to Forecasting Financial Market Volatility
- Elliott, Timmermann — Handbook of Economic Forecasting

References

- Mills, Markellos — The econometric modelling of financial time series
- Webster — Handbook of Price Impact Modeling

Tier 3 — Probability, Statistics & Causal Thinking

Core

- Wasserman — All of Statistics
- Gelman, Hill, Venkatani — Regression and Other Stories
- Gelman — Bayesian Data Analysis
- Blitzstein, Hwang — Introduction to Probability
- Jaynes — Probability theory_ the logic of science

Depth

- Billingsley — Probability and Measure
- Grimmert — probability and random processes
- Rosenthal — A First Look at Rigorous Probability Theory, Second Edition
- Bernardo, Smith — Bayesian Theory
- Efron, Hastie — Computer Age Statistical Inference_ Algorithms, Evidence, and Data Science
- Peters, Janzing, Schölkopf — Elements of Causal Inference
- Wainwright — High-dimensional statistics_ a non-asymptotic viewpoint
- Vershynin — High Dimensional Probability
- Williams, Rasmussen — Gaussian processes for machine learning
- der Vaart, Wellner — Weak Convergence and Empirical Processes_ With Applications to Statistics

Tier 4 — Machine Learning

Core

- Hastie, Tibshirani, Friedman — The Elements of Statistical Learning _ Data Mining, Inference and Prediction
- Murphy — Machine Learning_ A Probabilistic Perspective
- James, Witten, Hastie, Tibshirani — An Introduction to Statistical Learning - with Applications in Python
- Goodfellow, Bengio, Courville — Deep Learning
- Deisenroth, Faisal, Ong — Mathematics for Machine Learning
- De Prado — Advances in Financial Machine Learning
- de Prado — Machine Learning for Asset Managers
- Dixon, Halperin, Bilokon — Machine Learning in Finance_ From Theory to Practice

Depth

- Bishop — Pattern Recognition and Machine Learning
- Bishop, Bishop — Deep Learning_ Foundations and Concepts
- Geron — Hands-On Machine Learning with Skikit-Learn, Keras, and TensorFlow ...
- Prince — Understanding Deep Learning
- Schapire, Freund — Boosting_ Foundations and Algorithms
- Sutton, Barto — Reinforcement Learning_ An Introduction
- Bertsekas — a course in reinforcement learning
- Kochenderfer, Wheeler — Algorithms for optimization (ML-adjacent)
- Jordan — Introduction to Probabilistic Graphical Models (pack)
- Koller, Friedman — Probabilistic Graphical Models_ Principles and Techniques
- Williams, Rasmussen — Gaussian processes for machine learning
- Manokhin — Practical Guide to Applied Conformal Prediction in Python

References/Practice

- Janssen — Machine Learning for Algorithmic Trading_ Predictive models ...
- Kelly — Financial Machine Learning
- Coqueret, Guida — Machine Learning for Factor Investing_ Python Version
- Banachewicz, Massaron — The Kaggle Book ...
- Hudson & Thames — Guide to Modern Portfolio Optimization

Tier 5 — Optimization & Numerical Methods

Core

- Boyd — Convex Optimization
- Bertsekas — Nonlinear Programming
- Nocedal, Wright — Numerical optimization
- Sundaram — A First Course in Optimization Theory
- Ben-Tal, El Ghaoui, Nemirovski — Robust optimization

Depth

- Luenberger — Optimization by vector space methods
- Lan — First-Order and Stochastic Optimization Methods for Machine Learning
- Nesterov — Lectures on Convex Optimization
- Hazan — Introduction to Online Convex Optimization
- Catalfano, Chouai — Optimization Models
- Kochenderfer, Wheeler, Wray — Algorithms for decision making
- Mosek — MOSEKPortfolioCookbook-letter; MOSEKModelingCookbook-letter

Linear Algebra & Numerics (quick refs)

- Golub, Van Loan — Matrix Computations
- Trefethen — Numerical linear algebra
- Strang — Linear Algebra and Learning from Data; Linear Algebra and Its Applications
- Axler — Linear Algebra Done Right
- Meyer — Matrix analysis and applied linear algebra
- Bhatia — Matrix Analysis

Tier 6 — Asset Class Modules

A. Rates & Inflation

Core

- Tuckman — Fixed Income Securities-Tools for Today's Markets
- Andersen, Piterberg — Interest Rate Modeling
- Brigo, Mercurio — Interest Rate Models — Theory and Practice
- Burghardt — The Treasury Bond Basis-An in-Depth Analysis for Hedgers, Speculators, and Arbitrageurs
- Stigum, Crescenzi — Stigum's Money Market
- Homer, Leibowitz — Inside the Yield Book_ ... Third Edition
- Deacon, Derry, Mirfendereski — Inflation-indexed Securities
- James, Leister, Rieger — Inflation-Linked Bonds and Derivatives_ ...

Depth

- Lesniewski — Interest rate and credit models
- Richardson — Systematic Fixed Income_ An Investor's Guide
- Veronesi — handbook-of-fixed-income-securities
- Huggins, Schaller — SOFR Futures and Options_ A Practitioner's Guide
- Aikin — STIR Futures_ Trading Euribor and Eurodollar futures

B. FX & Cross-Currency

Core

- Weithers — Foreign exchange_ a practical guide to the FX markets
- Wystup — FX options and structured products
- Clark — Foreign Exchange Option Pricing_ A Practitioner's Guide
- DeRosa — Currency derivatives
- Iqbal — Foreign Exchange_ Practical Asset Pricing and Macroeconomic Theory

Depth

- James, Fullwood, Billington — FX Option Performance_ ...
- Jewitt — FX Derivatives Trader School
- Wiseman — Pricing Money

C. Commodities & Energy

Core

- Eydeland, Wolyniec — Energy and Power Risk Management
- McNally — Crude Volatility_ The History and the Future of Boom-Bust Oil Prices
- Downey — Oil 101
- Sturm — Trading Natural Gas_ Cash, Futures, Options and Swaps
- Thompson — Base metals handbook
- Bouchouev — Virtual Barrels

Depth

- Alizadeh, Nomikos — Shipping Derivatives and Risk Management
- Crabbe — Metals trading handbook
- Gottheif — Precious Metals Trading
- Kingsman — Out of the Shadows_ The New Merchants of Grain
- Morgan — Merchants of Grain
- Blas, Farchy — World For Sale_ Money, Power, And The Traders Who Barter The Earth's Resources
- Copetas — Metal men_ Marc Rich and the 10-billion-dollar scam
- Annmann — The King of Oil_ The Secret Lives of Marc Rich
- Vergin — The Prize

D. Equity Index & Trend/CTA

Core

- Clenow — Following the Trend_ Diversified Managed Futures Trading
- Greyserman, Kaminski — Trend following with managed futures _ the search for crisis alpha
- Cowel — Trend Following_ How to Make a Fortune in Bull, Bear and Black Swan Markets
- Ang — Asset Management_ A Systematic Approach to Factor Investing

Depth

- Tortoriello — Quantitative Strategies for Achieving Alpha
- Qian, Hua, Sorensen — Quantitative Equity Portfolio Management_ Modern Techniques and Applications
- Chincarinì — Quantitative Equity Portfolio Management

E. Credit & Structured

Core

- Dor, Desclee, Dynkin, Hyman, Polbennikov — Systematic Investing in Credit
- Gatto — The Credit Investor's Handbook_ Leveraged Loans, High Yield
- Barrieu, Albertini — The Handbook of Insurance-Linked Securities

Depth

- Schofield, Bowler — Trading the Fixed Income, Inflation and Credit Markets_ A Relative Value Guide
- Meissner — Correlation Risk Modeling and Management

Tier 7 — Derivatives, Volatility & Microstructure

Volatility & Options (Macro hedging)

- Gatheral — The Volatility Surface_ A Practitioner's Guide
- Derman — The Volatility Smile
- Natenberg — Option Volatility and Pricing_ Advanced Trading Strategies and Techniques
- Taleb — Dynamic Hedging_ Managing Vanilla and Exotic Options
- Sircidair — Volatility Trading; Option Trading_ Pricing and Volatility Strategies and Techniques; Positional Option Trading_ An Advanced Guide
- Bennett — Trading Volatility, correlation, term structure, and skew
- Bossu — Advanced Equity Derivatives_ Volatility and Correlation
- Haug — The Complete Guide to Option Pricing Formulas, 2nd ed
- Poon — A Practical Guide to Forecasting Financial Market Volatility
- GS QIS notes (entire folder)

Microstructure & Execution

- Harris — Trading and Exchanges_ Market Microstructure for Practitioners
- O'Hara — Market Microstructure Theory
- Hasbrouck — Empirical Market Microstructure_ The Institutions, Economics, and Econometrics of Securities Trading
- Bouchaud, Bonart, Donier, Gould — Trades, Quotes and Prices
- Lehalle, Laruelle — Market Microstructure in Practice

HFT & Market Structure (context)

- Aldridge — High-frequency trading_ a practical guide to algorithmic strategies and trading systems
- Durbin — All About High-Frequency Trading
- Cartea — Algorithmic and High-Frequency Trading
- Florescu, Mariani, Stanley, Vens — Handbook of High-Frequency Trading and Modeling in Finance
- Bodek — The Problem of HFT ...
- Amuk, Saluzzi — Broken Markets_ How High Frequency Trading and Predatory Practices ...
- MacKenzie — Trading at the Speed of Light_ How Ultrafast Algorithms Are Transforming Financial Markets
- Patterson — Dark Pools_ The Rise of the Machine Traders and the Rigging of the U.S. Stock Market
- Lewis — Flash Boys_ A Wall Street Revolt; Kovac — Flash Boys_ Not So Fast_ ...; Vaughan — Flash Crash_ ...

Tier 8 — Portfolio Construction, Risk & Attribution

Core

- Meucci — Risk and Asset Allocation
- Grinold, Kahn — Active Portfolio Management_ A Quantitative Approach for Producing Superior Returns and Controlling Risk
- Grinold, Kahn — Advances in Active Portfolio Management_ New Developments in Quantitative Investing
- Scherer — Portfolio Construction and Risk Budgeting
- Guerard — Portfolio construction, measurement and efficiency
- Connor, Goldberg, Korajczyk — Portfolio Risk Analysis

Depth

- Alonso, Calamarena, Guerrero — Quantitative Portfolio Optimization _ Advanced Techniques and Applications
- Pav — The Sharpe Ratio_ Statistics and Applications
- Satchell — Asset Management
- Paleologo — Advanced Portfolio Management_ A Quant's Guide for Fundamental Investors; The Elements of Quantitative Investing
- Ang — Asset Management_ A Systematic Approach to Factor Investing (again)

Tier 9 — Systematic Trading

Core

- Chan — Algorithmic Trading; Quantitative trading _ how to build your own algorithmic trading business
- Halls-Moore — Successful Algorithmic Trading
- Kissell, Moore — Optimal trading strategies_ quantitative approaches for managing market impact and trading risk
- Kissell — Algorithmic Trading Methods_ Applications Using Advanced Statistics, Optimization, and Machine Learning Techniques
- Tulchinsky et al. — Finding Alphas - A Quantitative Approach to Building Trading Strategies
- Kakushadze — 151 Trading Strategies
- Pardo — The Evaluation and Optimization of Trading Strategies
- Kaufman — Trading Systems and Methods
- Pole — Statistical Arbitrage_ Algorithmic Trading Insights and Techniques
- Higgs — Python for Algorithmic Trading_ From Idea to Cloud Deployment
- Moore — Advanced Algorithmic Trading

Depth

- Carver — Systematic Trading_ A unique new method for designing trading and investing systems; Smart portfolios; Leveraged Trading_ ...
- Halls-Moore, Huggins, Schaller — Fixed Income Relative Value Analysis, A Practitioner's Guide ...
- Richardson — Systematic Fixed Income_ An Investor's Guide (again)
- Rubano — Trader Construction Kit
- Donnelly — Alpha Trader_ The Mindset, Methodology and Mathematics of Professional Trading

Practice/Interviewing

- Zhou — A Practical Guide to Quant Finance Interviews
- Stefanica — 150 Most Frequently Asked Questions on Quant Interviews
- Joshi — Quant Job Interview Questions And Answers

Tier 10 — Data Engineering & Software

Core

- Kleppmann — Designing data-intensive applications_ ...
- Reis, Housley — Fundamentals of Data Engineering_ Plan and Build Robust Data Systems
- Crickard — Data Engineering with Python_ ...
- Richards, Ford — Fundamentals of Software Architecture_ An Engineering Approach
- Burns — Designing_Distributed_Systems

Python & Performance

- Ramalho — Fluent Python, 2nd Edition
- Gorelick, Ozsvald — High Performance Python_ Practical Performant Programming for Humans
- Perovial, Gregory — Architecture Patterns and Patterns in Random Matrix Theory
- Balaraman, Ballabio — QuantLib Python Cookbook

Signal Processing & Systems (for macro data)

- Oppenheim — Signals, systems & inference
- Anand — Digital Signal Processing An Introduction
- Stanford — EE269 - Signal Processing for Machine Learning (pack)

Tier 11 — Math Foundations

- Strang — Differential Equations and Linear Algebra; Computational Science and Engineering
- Axler — Measure, Integration and Real Analysis
- Tao — An Introduction to Measure Theory; Topics in Random Matrix Theory
- Karatzas, Shreve — Brownian Motion and Stochastic Calculus
- Oksendal — Stochastic Differential Equations_ An Introduction with Applications
- Steele — Stochastic Calculus and Financial Applications
- van Handel — Stochastic Calculus, Filtering, and Stochastic Control
- Lawler — Stochastic Calculus_ An Introduction with Applications
- Billingsley — Probability and Measure (again)
- Golub, Van Loan — Matrix Computations (again)
- Meyer — Matrix analysis and applied linear algebra (again)
- Wainwright, Jordan — Graphical Models, Exponential Families, and Variational Inference
- Cover, Thomas — Elements of Information Theory
- Mézard, Montanari — Information, Physics, and Computation